



EE5060

Sensors and Instrumentation for Automation

EE5111

Selected Topics in Industrial Control & Instrumentation

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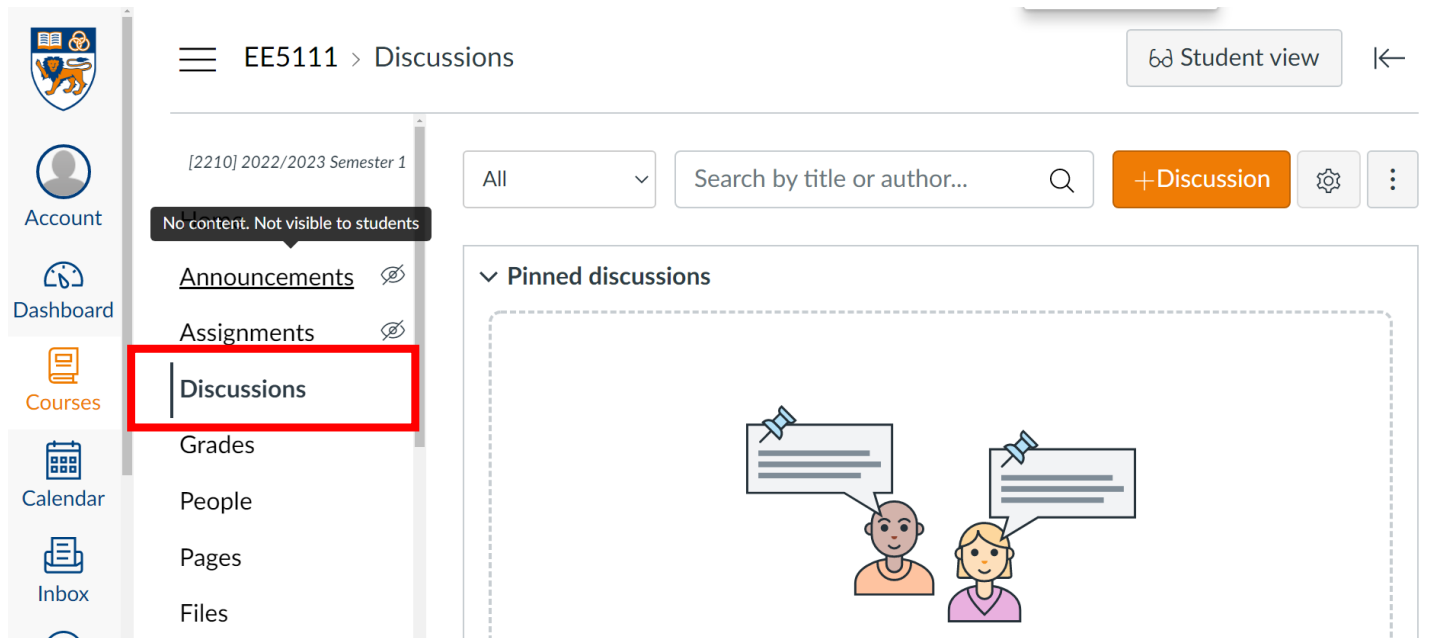
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- NUS Canvas
<https://canvas.nus.edu.sg/>
 - Maintain communication after class
 - A place for you to discuss anytime, anywhere, and to exchange information



Your feedback is valuable to us...

- Join the online student feedback exercise!
- Feedback report includes two parts:
 - Course report and teacher report
- **Course report**
 - Overall opinion, your expected grade, difficulty level
- **Teacher report**
 - Feedback to each lecturer

Question
The teacher has enhanced my thinking ability.
The teacher provided timely and useful feedback.
The teacher has increased my interest in the subject.

Leave your (detailed) comments, not just scores.

2025 Student Feedback Review

- High response ratio last year

Course	EE5060 - SENSORS AND INSTRUMENTATION FOR AUTOMATION
Academic Year/Sem	2025/2026 - Sem 1
Department	ELECTRICAL & COMPUTER ENGINEERING
Faculty	COLLEGE OF DESIGN & ENGINEERING

Note: Class Size = Invited; Response Size = Responded; Response Rate = Response Ratio

Raters	Student
Responded	17
Invited	26
Response Ratio	65%

Course	EE5111 - SELECTED TOPICS IN INDUSTRIAL CONTROL & INSTRUMENTATION
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Note: Class Size = Invited; Response Size = Responded; Response Rate = Response Ratio

Raters	Student
Responded	40
Invited	55
Response Ratio	73%

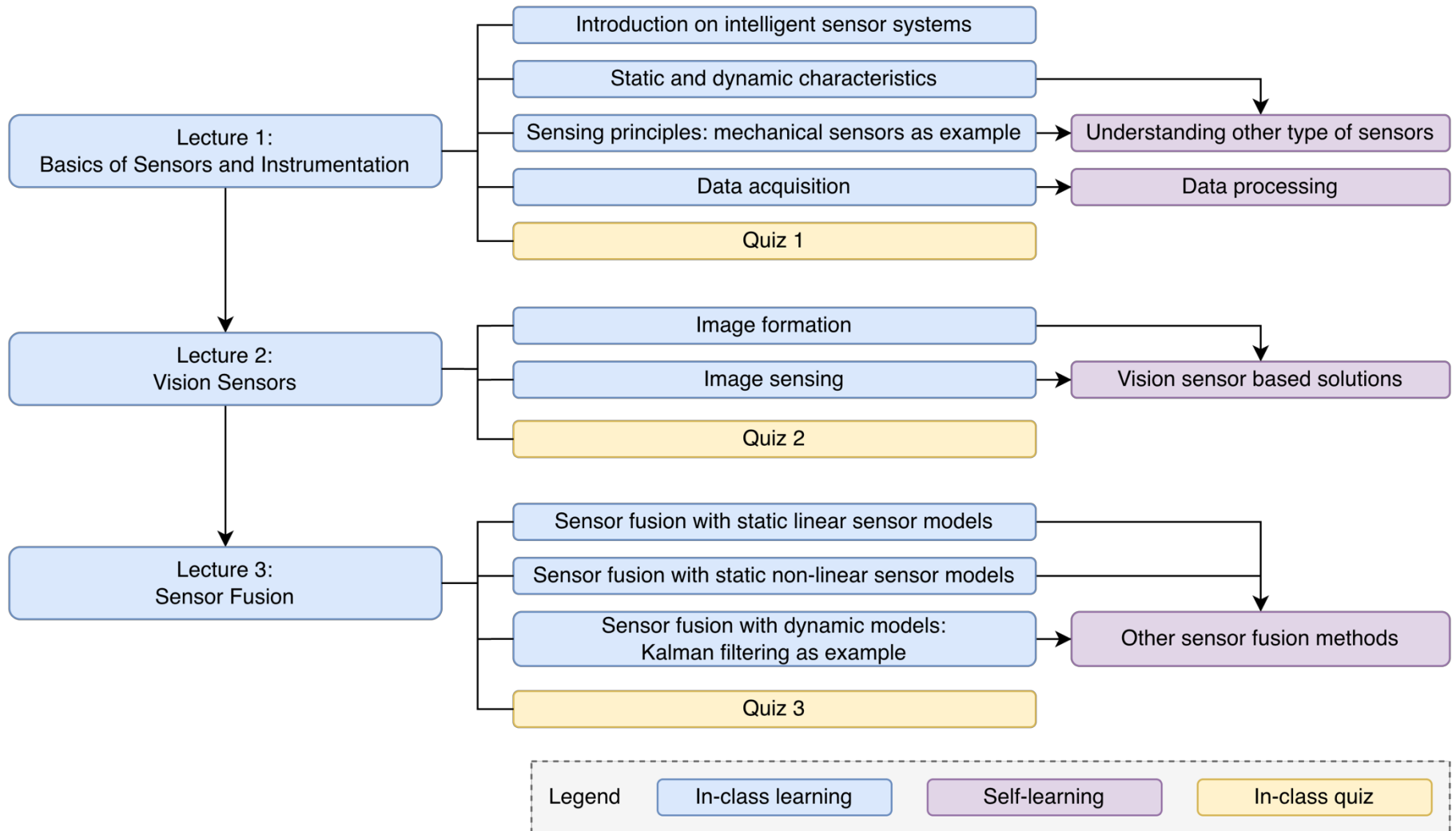
- Change:
Project-based CA (last year) → In-class quizzes (this year)

Continuous Assessment (CA)

- **In-class Quizzes for Lectures 1–3**

- One quiz will be given immediately after each lecture.
- Each quiz: 25 marks, about 15 minutes.
- Your CA mark will be the average of your **best two** quiz scores out of three.
- Questions are based on the concepts and examples discussed in class.
- Focus: **conceptual understanding and practical reasoning, not lengthy calculations.**

Scope of the Lectures 1 to 3



Outline

Introduction

- Intelligent sensor systems

Basics of Sensors and Instrumentation

- Static characteristics and dynamic characteristics
- Basics of sensing principles: mechanical sensors as examples
- Data acquisition

Learning Objectives

- In this lecture, you are expected to:
 1. Describe and explain the basic concepts in sensing and instrumentation;
 2. Understand the **characteristics** and **characterization** of sensors;
 3. Understand the interfacing and data acquisition of sensors.

Systems, Sensors, and Intelligence

- **System**

- A combination of two or more elements, subsystems and parts necessary to carry out one or more functions
- To interact with the real world, a system requires:
 - 1) **sensors** as inputs devices, 2) **actuators** as output devices, and 3) **processing**

- **Sensor**

- Device that receives and responds to a stimulus

Stimulus: mechanical, thermal, magnetic, electric, optical, chemical...

Response: an electrical signal (in most cases)

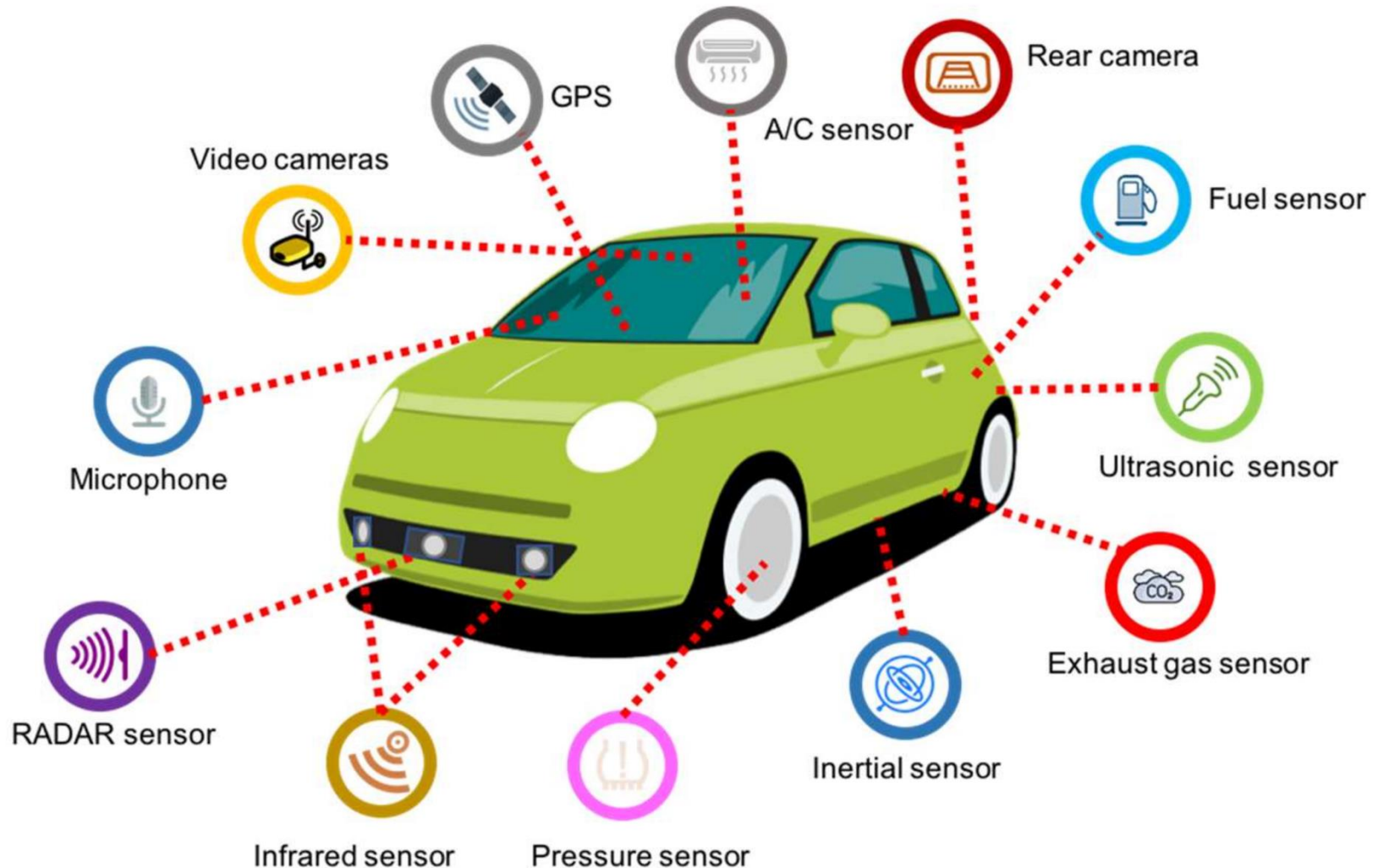
- **Intelligence**

- The ability to combine **a priori knowledge** and **measurement**
- **Decision-making / reasoning** abilities

Together, these form the basis of an intelligent sensor system.

Intelligent Sensor Systems (ISS): Example

- **Autonomous vehicles**



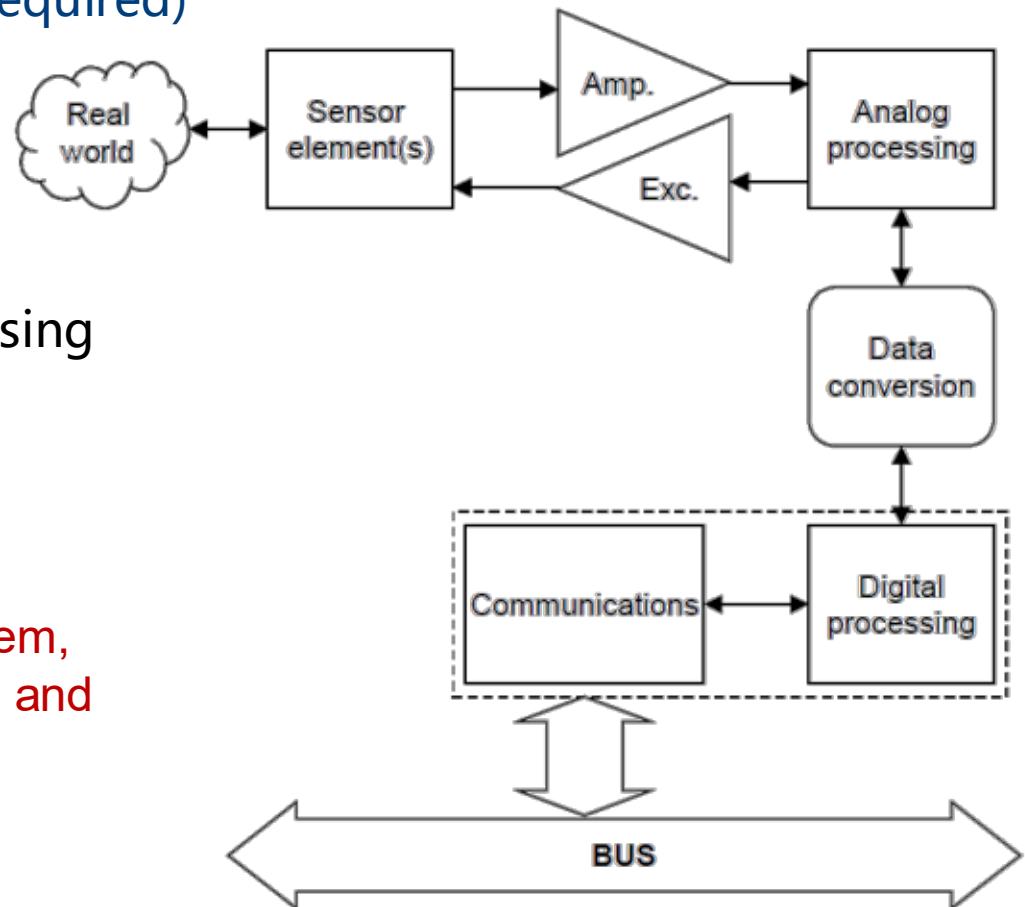
Intelligent Sensor Systems (ISS): Example

- **Your smart phone**



Building Blocks of ISS

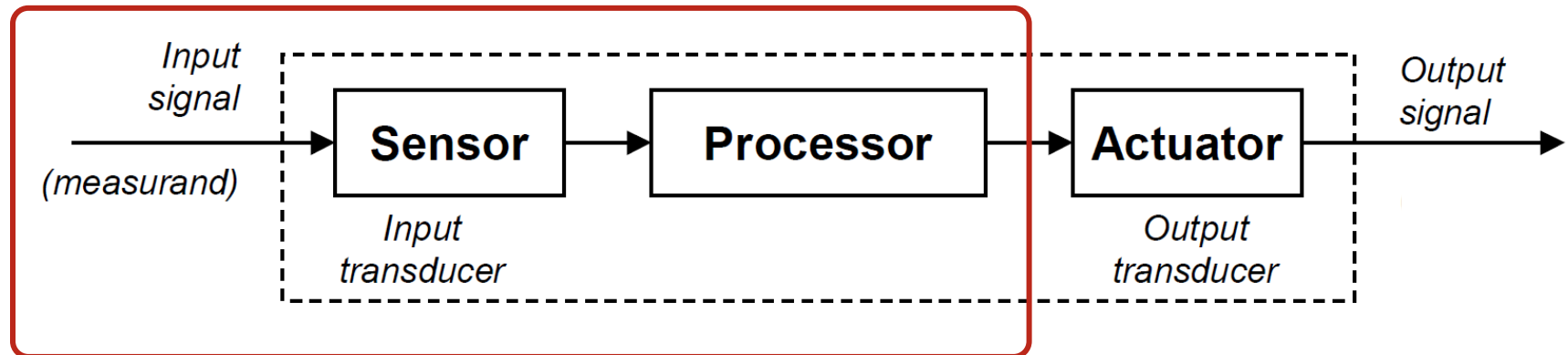
- The principal sub-systems within an ISS are
 - Primary sensing element(s)
 - Sensor bias/excitation (if required)
 - Amplification
 - Analogue processing
 - DAC
 - Digital information processing
 - Communications



Can we simplify the above system, especially when doing research and prototype development?

Transducers: Sensors and Actuators

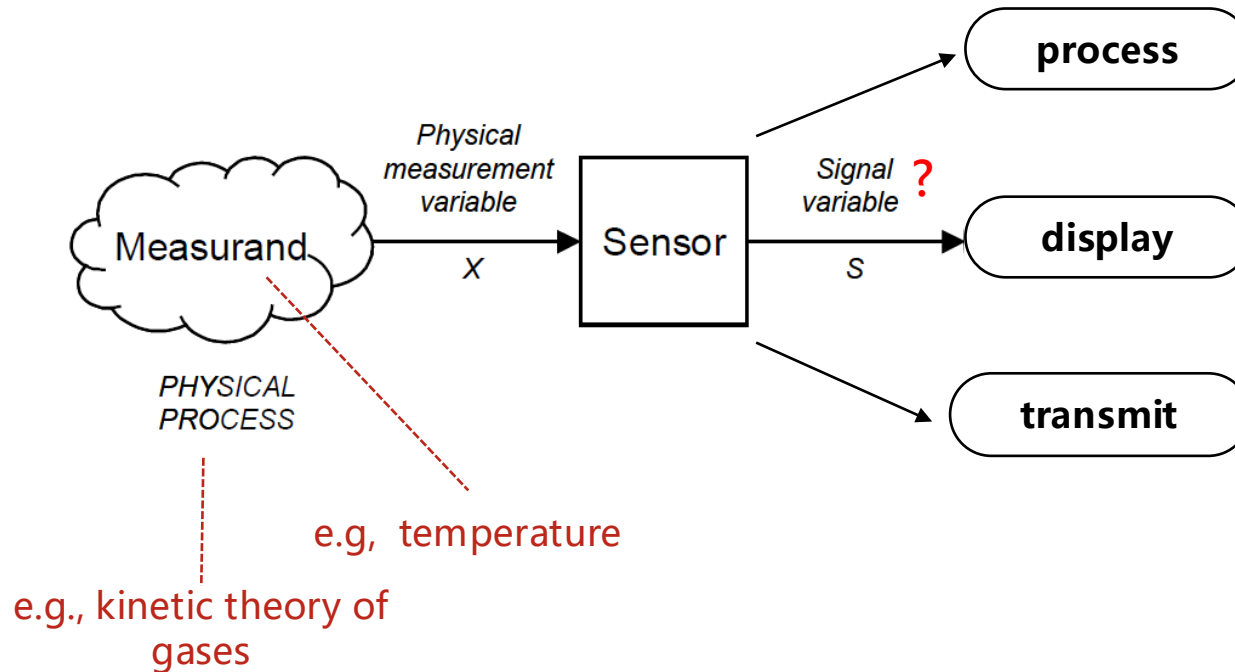
- **Transducer:** ^{换能器} A device that converts a signal from one physical form to a corresponding signal having a different physical form
 - Sensor: an input transducer (e.g., a microphone)
 - Actuator: an output transducer (e.g., a loudspeaker)



Simplified ISS

Measurement Process

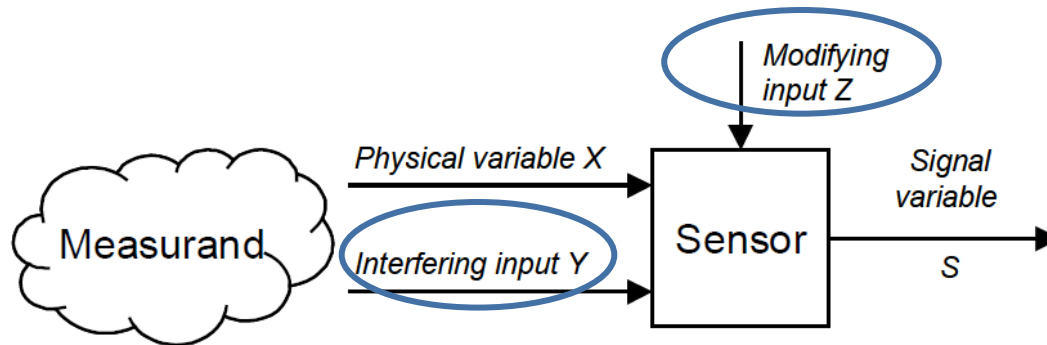
- A **simple** instrument model



There is no perfect model!

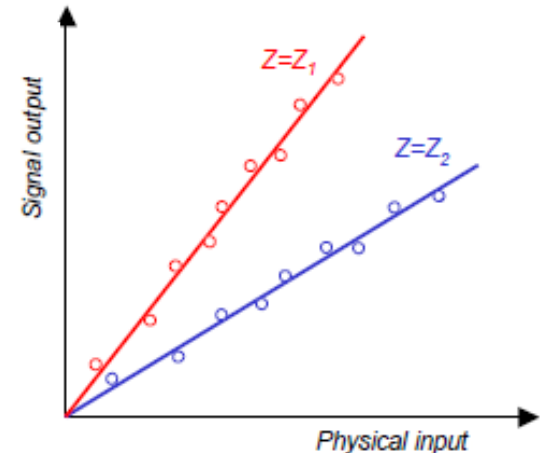
Measurement Process

- A **practical** instrument model



- **Interfering inputs (Y):**
Quantities to which the instrument is unintentionally sensitive
- **Modifying inputs (Z):**
Quantities that cause a change in the input-output relations of the instrument

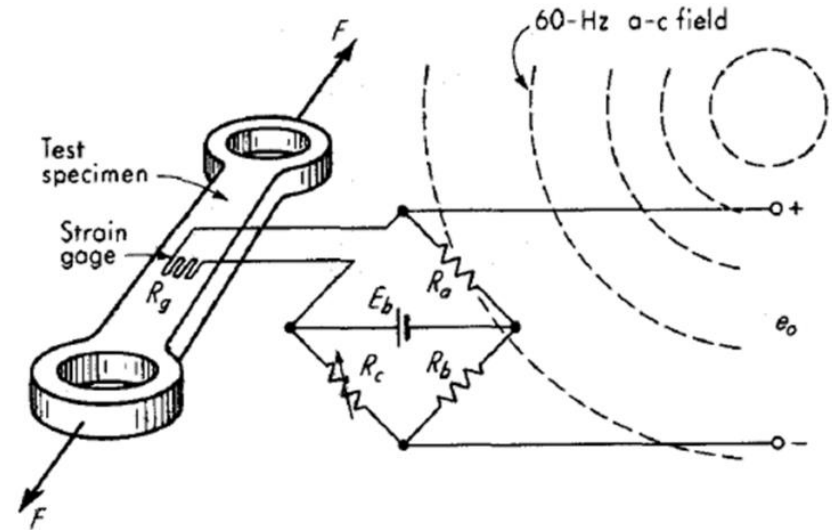
Input-Output Curve



Examples: Interfering/Modifying Inputs

- Interfering inputs

- The 60-Hz magnetic field induces voltages in the strain-gauge circuit.
- The gauge temperature 应变片 causes a change in gauge resistance; cause a voltage output even if there is no strain.



Good or bad?

- Modifying inputs

- The gauge factor GF is sensitive to temperature
- The battery voltage E_b also changes the proportionality factor

$$\Delta R_g = (GF)R_g\epsilon$$

$\Delta R_g \triangleq$ change in gage resistance, Ω^*

$GF \triangleq$ gage factor, dimensionless

$R_g \triangleq$ gage resistance when unstrained, Ω

$\epsilon \triangleq$ unit strain, cm/cm

$$e_o = -(GF)R_g\epsilon E_b \frac{R_a}{(R_g + R_a)^2}$$

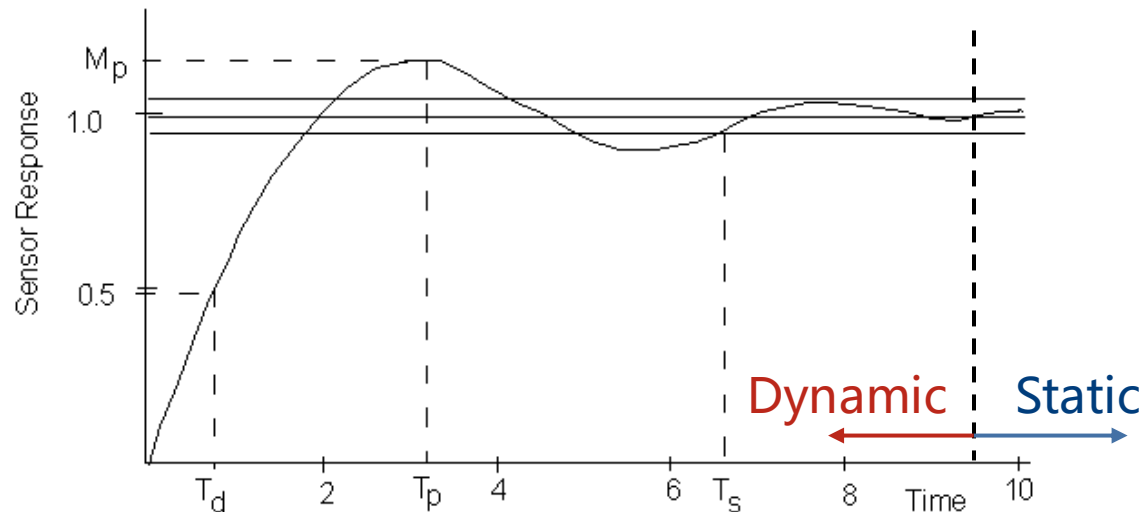
Sensor Characteristics

- **Static characteristics**

- The properties of the system after all transient effects have settled to their final or steady state

- **Dynamic characteristics**

- The properties of the system transient response to an input



Static characteristics: Accuracy, Discrimination

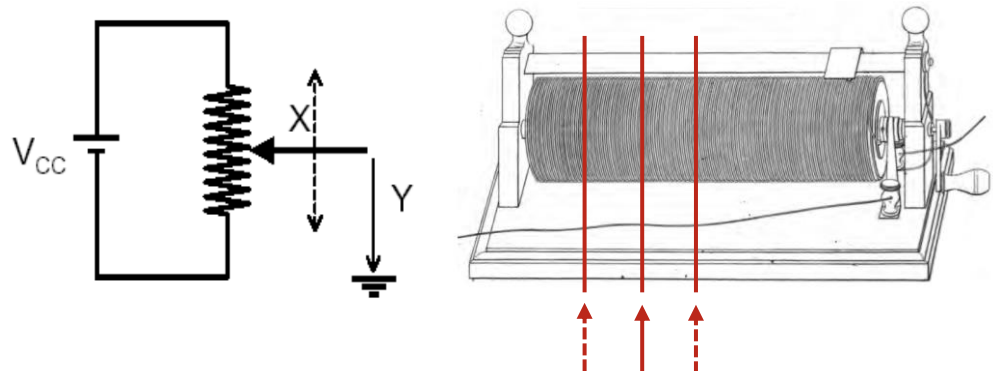
- **Accuracy** is the capacity of a measuring instrument to give RESULTS close to the TRUE VALUE of the measured quantity.

$$\text{ABSOLUTE ERROR} = \text{RESULT} - \text{TRUE VALUE}$$

$$\text{RELATIVE ERROR} = \frac{\text{ABSOLUTE ERROR}}{\text{TRUE VALUE}}$$

- **Discrimination/resolution** is the minimal change of the input necessary to produce a detectable change at the output.

Example: Potentiometer



Resolution limited by discrete coil turns

Systematic errors and random errors

- **Systematic errors**

- Examples: calibration errors, human observation errors (parallax errors), environmental errors (temperature)
- Systematic errors sometimes can be corrected with **COMPENSATION** methods

- **Random errors**

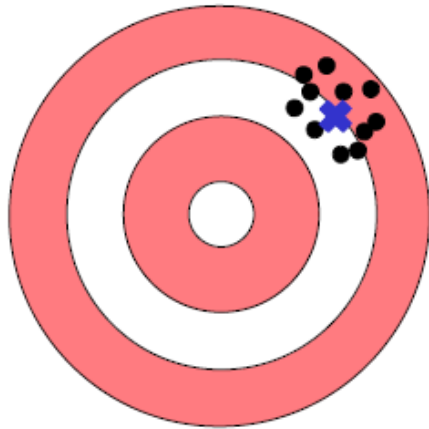
- Also called **NOISE**: a signal that carries no information
- True random errors (white noise) follow a **Gaussian distribution**
- **Random errors can only be reduced**

Static characteristics: Precision

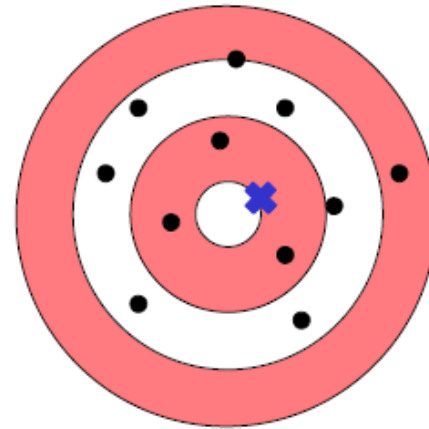
- **Precision:** The capacity of a measuring instrument to give the same reading when **repetitively** measuring the same quantity under the same prescribed conditions.

precision $\neq$ accuracy

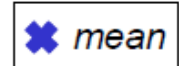
Which result is more accurate? Which result is more precise?



Result A



Result B



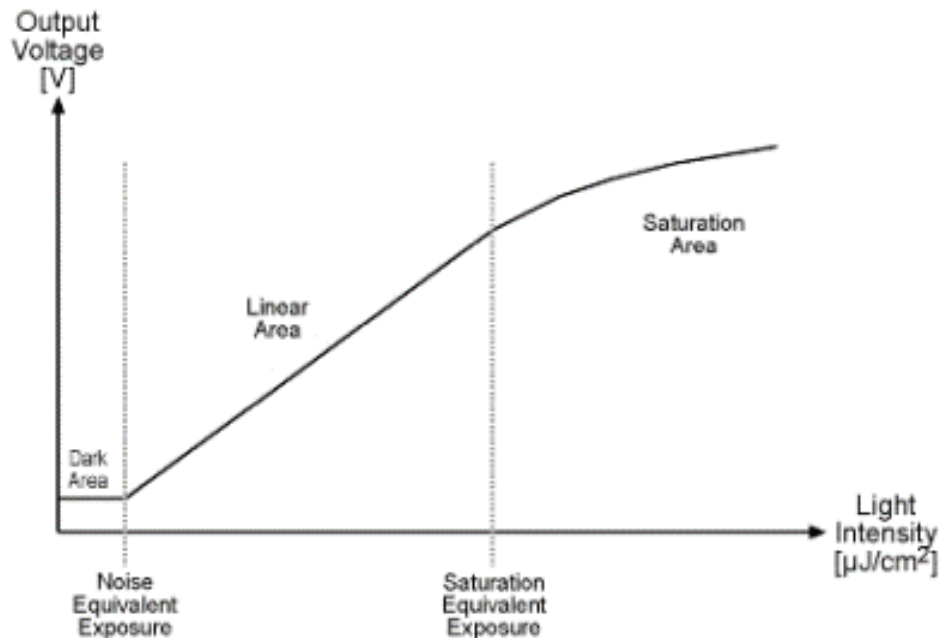
More Static Characteristics

- **Input range**

- The **maximum and minimum value** of the physical variable that can be measured (i.e., -40C/100C in a thermometer)
- Output range can be defined similarly

- **Sensitivity**

- The **slope** of the calibration curve $y=f(x)$



More Static Characteristics

- **Linearity**

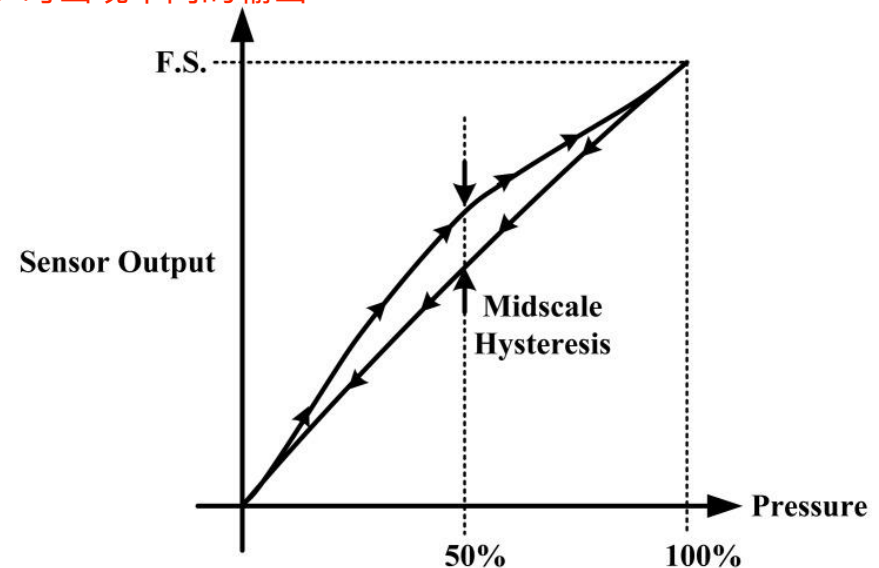
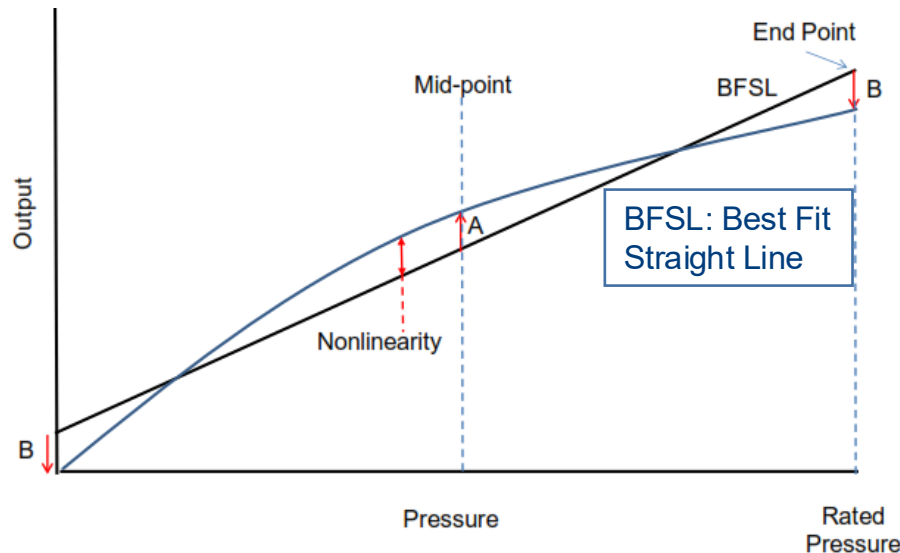
- The **closeness** of the calibration curve to a specified straight line (i.e., **theoretical behavior, least-squares fit**)

最小二乘法

- **Hysteresis** 迟滞

- The **difference** between two output values that correspond to **the same input** depending on the trajectory followed by the sensor

输入增加和输入减少时出现不同的输出



Exercise

1. What term describes the maximum expected error associated with a measurement or a sensor?

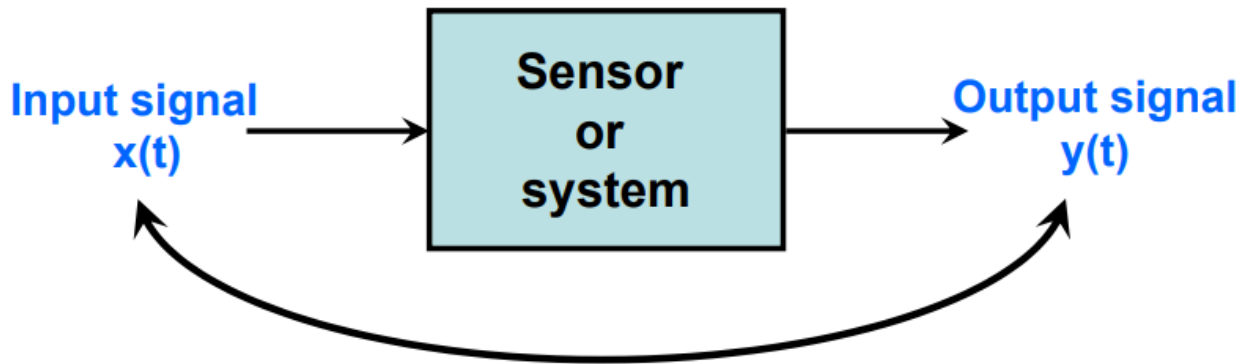
- a) Accuracy.
- b) Resolution.
- c) Precision.
- d) Range.

• **Correct Answers:**

Dynamic Characteristics

- How well a sensor respond to **changes** in its input?

Sensor or the measurement system must be able to respond fast enough to keep up with the input signals!



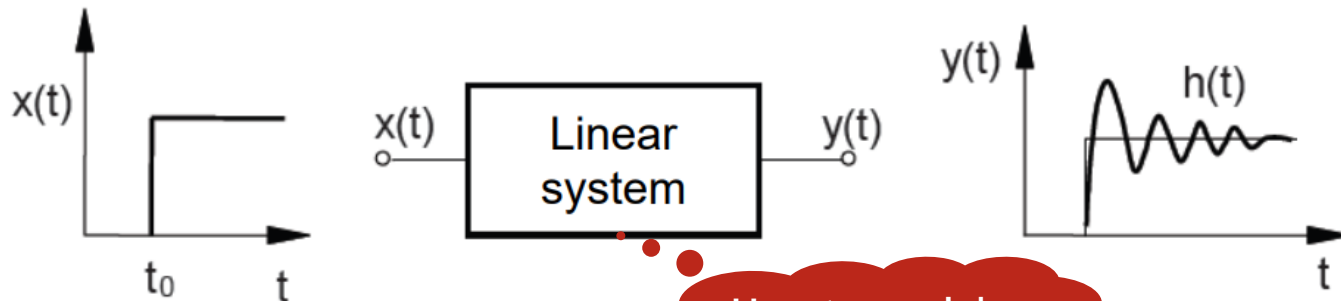
- To simplify the problem, consider linear area of the sensor.

We already have the results in **linear system theory**!

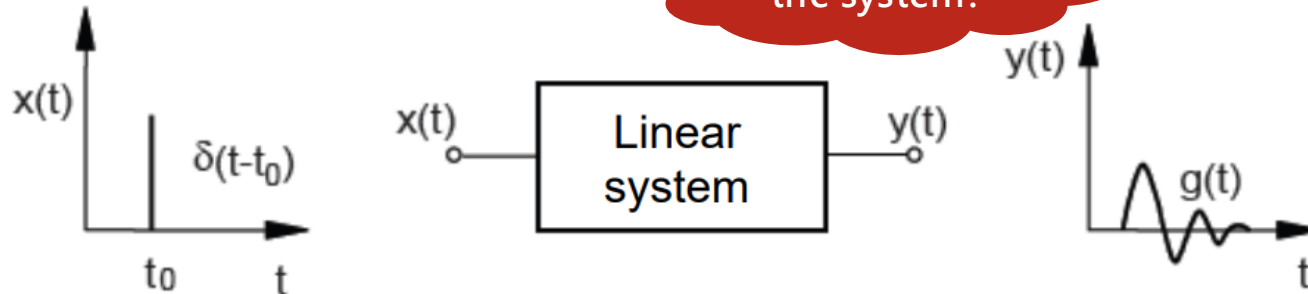
Dynamic Characteristics

- The commonly-used input signals are selected as follows:

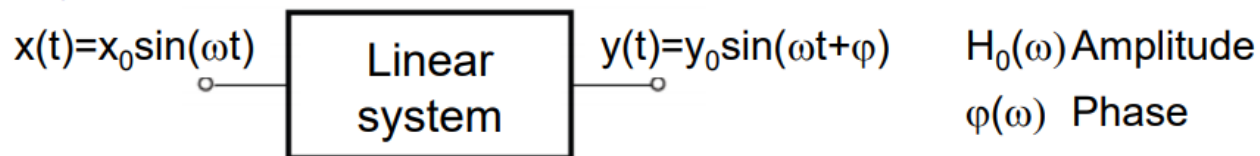
Step-Function



Impulse



Frequency excitation



How to model the system?

Dynamic Models

- The dynamic response of a sensor is typically approximated as linear and can be modelled using a simplified linear differential equation with constant coefficients.

$$a_k \frac{d^k y(t)}{dt^k} + \dots a_2 \frac{d^2 y(t)}{dt^2} + a_1 \frac{dy(t)}{dt} + a_0 y(t) = x(t)$$

- In practice, these models are confined to zero, first and second order. Higher order models are rarely applied.
- Applying the **Laplace transform** to the sensor model yields

$$\mathcal{L} \left[a_k \frac{d^k y}{dt^k} + \dots a_2 \frac{d^2 y}{dt^2} + a_1 \frac{dy}{dt} + a_0 y(t) = x(t) \right]$$

$\Downarrow$

$$(a_k s^k + \dots a_2 s^2 + a_1 s + a_0) Y(s) = X(s)$$

$\Downarrow$

Transfer function $G(s) = \frac{Y(s)}{X(s)} = \frac{1}{a_k s^k + \dots a_2 s^2 + a_1 s + a_0}$

Why use
Laplace
transform?

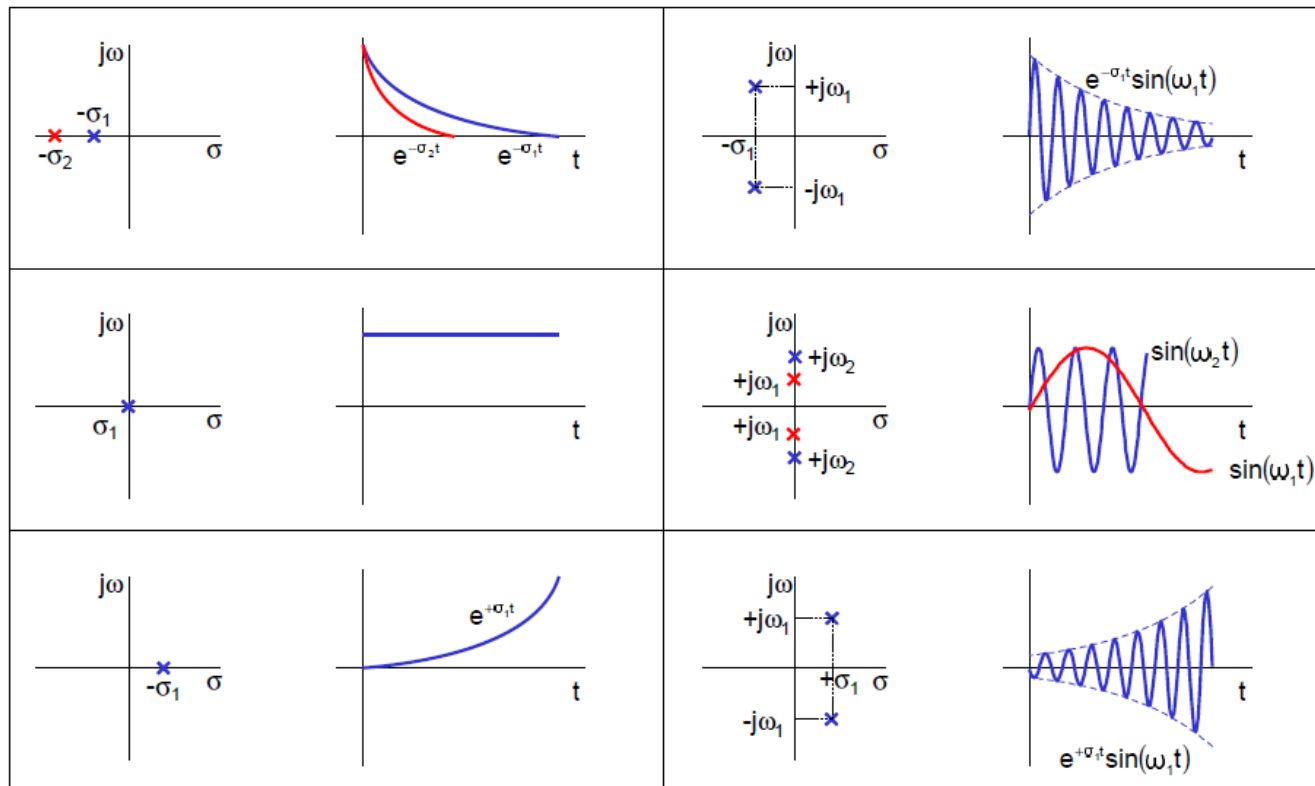
Differential
equations



Algebraic
equations

Pole Location and Dynamic Behavior

- The position of the **poles** of $G(s)$ - zeros of the denominator in the s-plane determines the dynamic behavior of the sensor such as: oscillating components, exponential decays, instability.



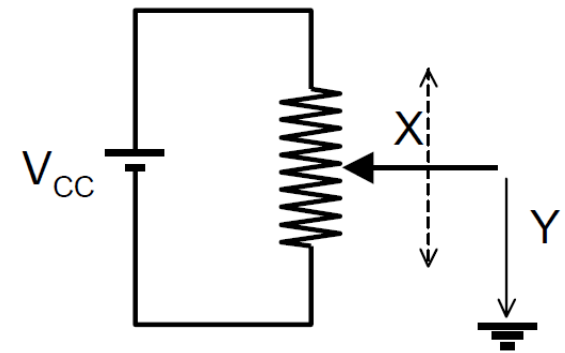
Pole locations determine the natural dynamic behavior of a sensor.

Zero-order Systems

- Input-output relation

$$y(t) = k \cdot x(t) \Rightarrow \frac{Y(s)}{X(s)} = k$$

- Zero-order is the desirable response of a sensor
 - No delays
 - Infinite bandwidth
 - The sensor only changes the amplitude of the input signal
- No energy-storing elements inside!
- Example: a potentiometer measuring **linear** and **rotary** displacements

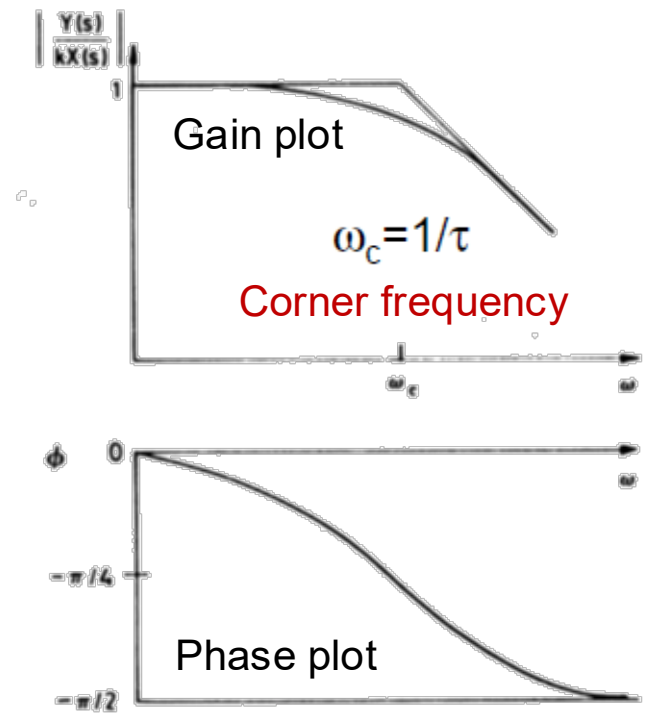
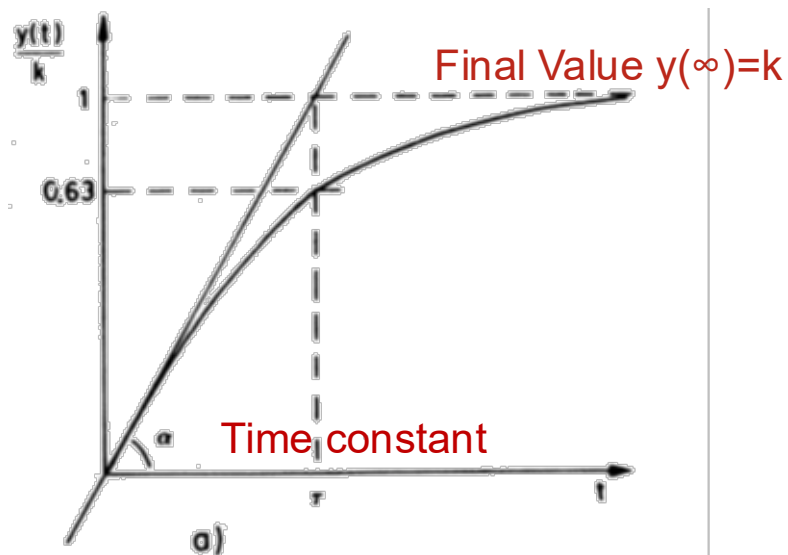


First-order Systems

- Input-output relation

$$a_1 \frac{dy}{dt} + a_0 y(t) = x(t) \Rightarrow \frac{Y(s)}{X(s)} = \frac{1}{a_1 s + a_0} = \frac{k}{\tau s + 1}$$

- First-order sensors have one element that stores energy and one that dissipates it.
- Step response and frequency response**



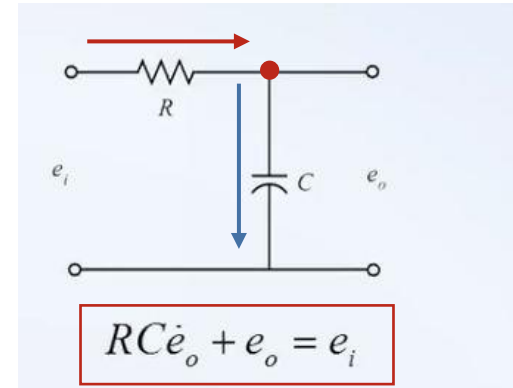
First-order Systems: Example

- RC circuit:

- $Q = CV$

- $I = dQ/dt$

Kirchhoff's current law

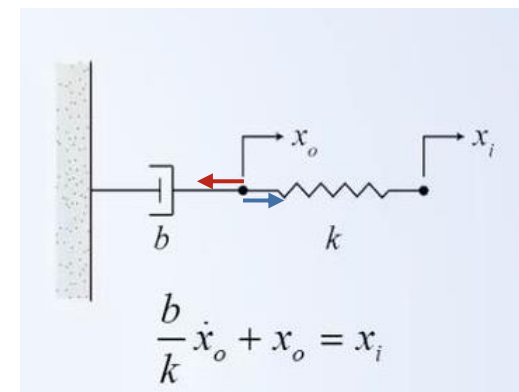


- Spring-damper:

- $F = -k \Delta x$

- $F = -bv$

Force equilibrium



Second-order Systems

- Input-output relation

$$a_2 \frac{d^2 y}{dt^2} + a_1 \frac{dy}{dt} + a_0 y(t) = x(t) \Rightarrow \frac{Y(s)}{X(s)} = \frac{1}{a_2 s^2 + a_1 s + a_0}$$

- Rewrite the transfer function as

$$\frac{Y(s)}{X(s)} = \frac{k\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\text{with } k = \frac{1}{a_0}, \zeta = \frac{a_1}{2\sqrt{a_0 a_2}}, \omega_n = \sqrt{\frac{a_0}{a_2}}$$

static
gain

zeta:
damping
coefficient

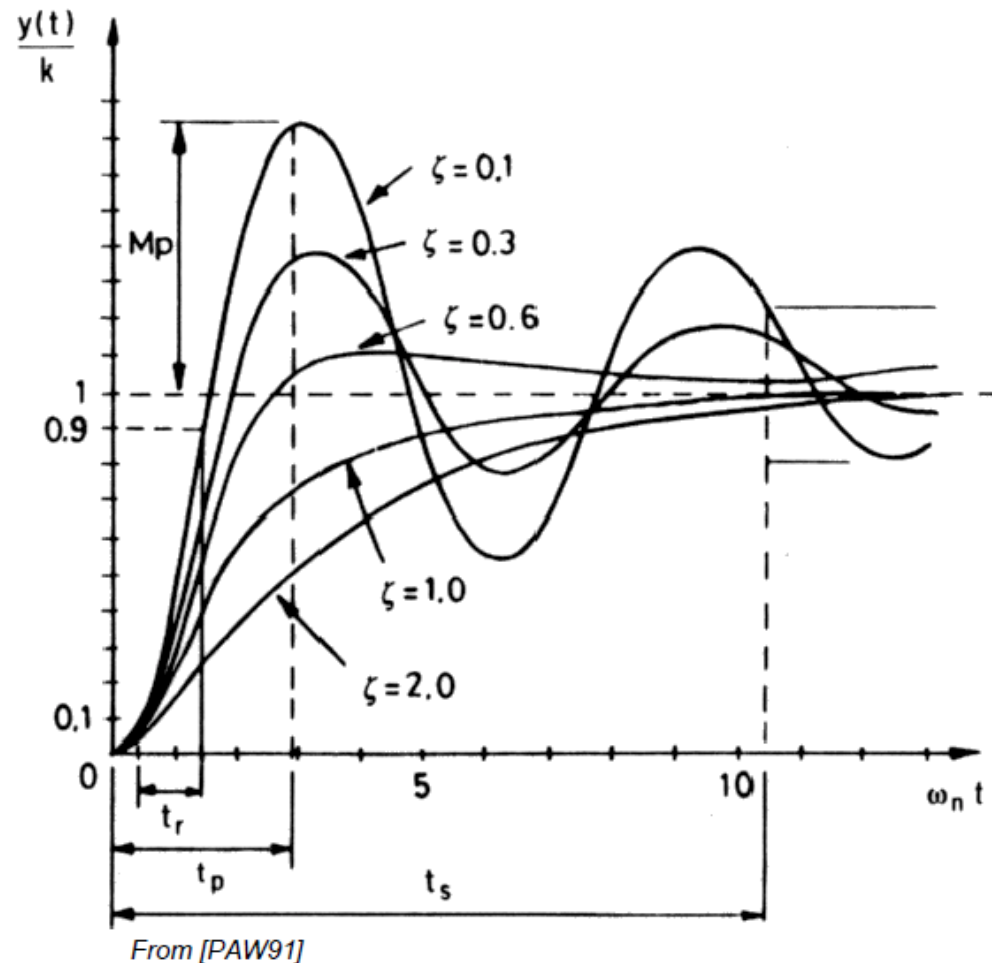
阻尼比

omega:
natural
frequency



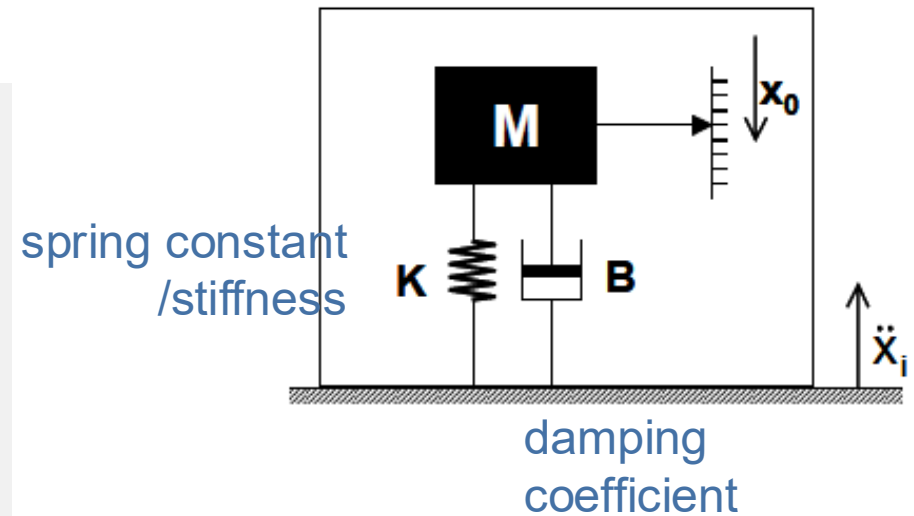
Second-order Step Response 阶跃响应

- Response types
 - Underdamped ($\zeta < 1$)
 - Critically damped ($\zeta = 1$)
 - Overdamped ($\zeta > 1$)
- Response parameters
 - Rise time (t_r)
 - Peak overshoot (M_p)
 - Time to peak (t_p)
 - Settling time (t_s): 5%/2%



Example of Second-order Systems

- Spring-mass-damper accelerometer
 - Analyse the object M based on Newton's law of motion



**When we say, “sensor A is an n -th order sensor,”
is that a fact or an assumption?”**

Engineering Practice: Parameter Estimation (Characterization)

- Background: How to make the datasheet of a sensor?

Electro-optical characteristics @ 23°C

Symbol	Characteristic	Test Condition	Min	Typ	Max	Unit
	Active area		diameter 230			μm
	Active area		0.04			mm^2
I_D	Dark current	$M = 100$		0.5	1	nA
C	Capacitance	$M = 100$; $f = 100 \text{ kHz}$		0.3		pF
	Responsivity	$M = 100$; $\lambda = 905 \text{ nm}$	52	58		A/W
t_R	Rise time	$M = 100$; $\lambda = 905 \text{ nm}$; $R_L = 50 \Omega$		0.5		ns
	Cut-off frequency	-3dB		0.6		GHz
V_{BR}	Breakdown voltage*	$I_R = 2 \mu\text{A}$	160		200	V
	Temperature coefficient	Change of V_{BR} with temperature	1.25	1.45	1.55	V/K
	Excess noise factor	$M = 100$; calculated		2.5		
	Excess noise index	$M = 100$; calculated		0.2		

* $\pm 2\text{V}$ measuring tolerance on upper and lower limits

Engineering Practice: Parameter Estimation (Characterization)

- Problem Statement:

Your company developed a new ammeter (electrical current  electrical voltage).

You are the engineer who oversees the product data sheet.

Engineering Practice: Parameter Estimation (Characterization)

Engineering Practice: Parameter Estimation (Characterization)

Summary - Introduction, Sensor Characteristics

- Introduction of Intelligent Sensor Systems
- Transducers, sensors and measurements
- Static sensor characteristics
- Dynamic sensor characteristics

Break

- Intelligent Vision Sensor with on-chip AI



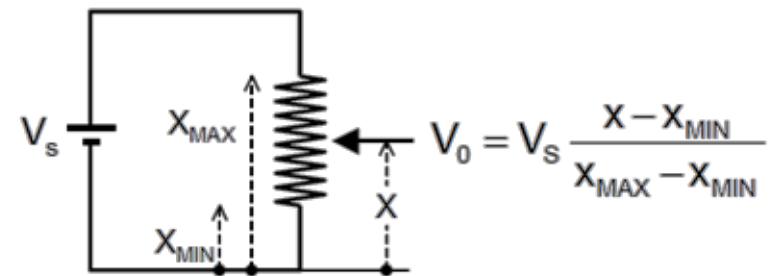
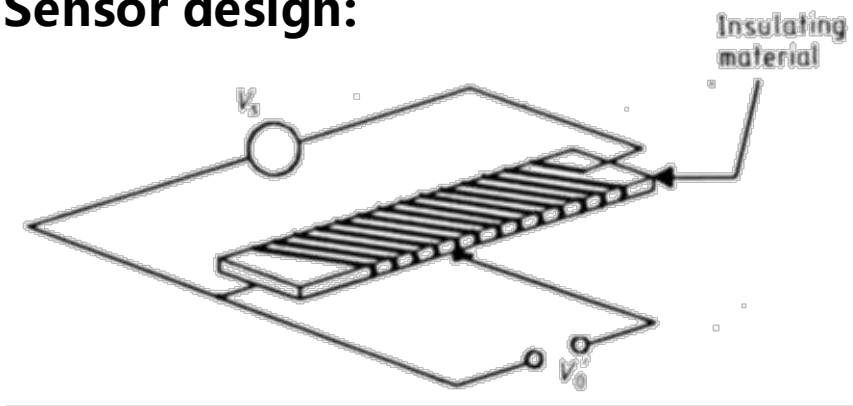
Sensing Principles: Mechanical Measurands

- Let us discuss sensors for mechanical measurands for example.
 - **Displacement measurement**
 - Resistive sensors
 - Capacitive sensors
 - Inductive sensors
 - **Force and acceleration measurement**
 - Strain gauges
 - Cantilever beam-based sensors

Resistive Displacement Sensors

- **Principle:** A resistance with a movable contact (a **potentiometer**) may be used to measure linear or rotational displacements
 - A known voltage is applied to the resistor ends
 - The output voltage at the contact is proportional to the displacement

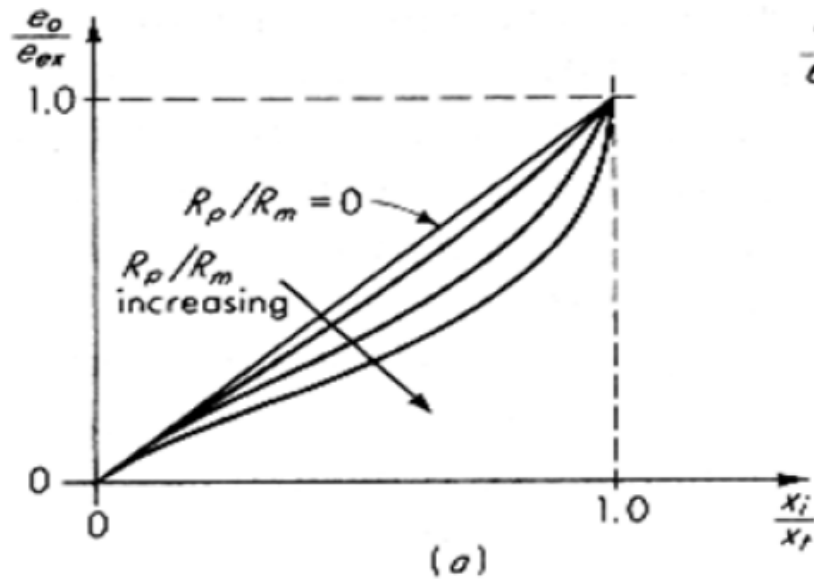
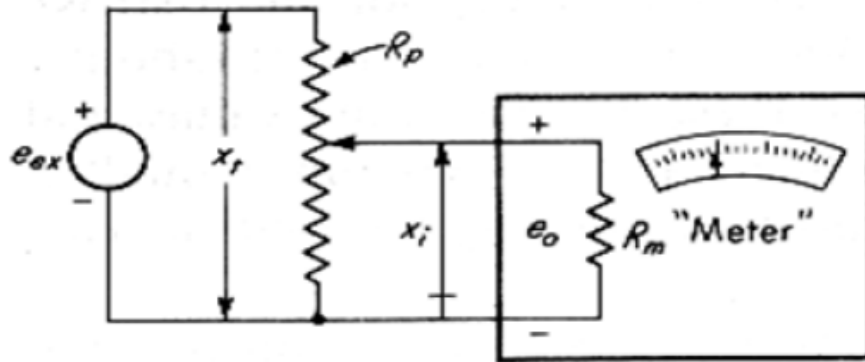
Sensor design:



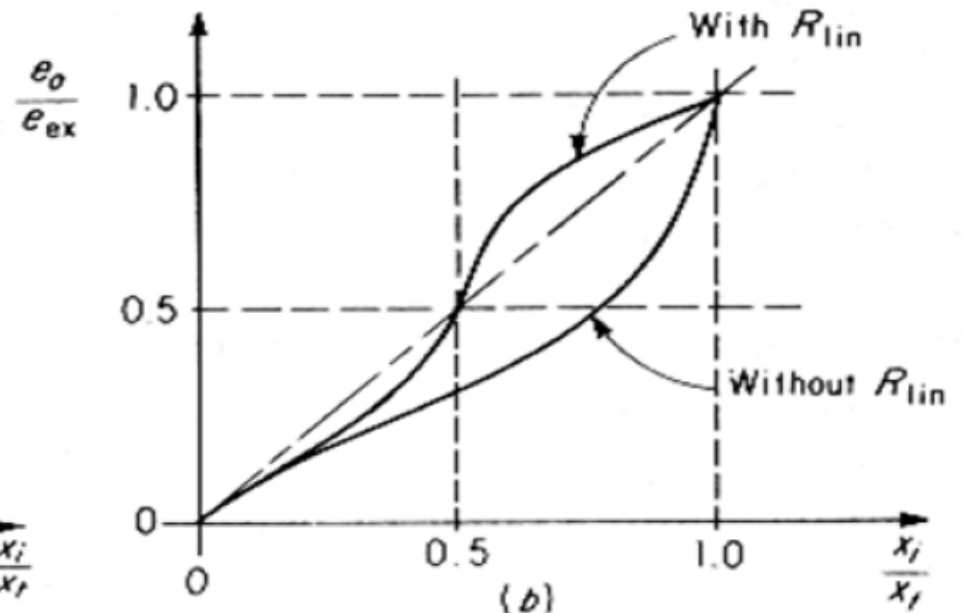
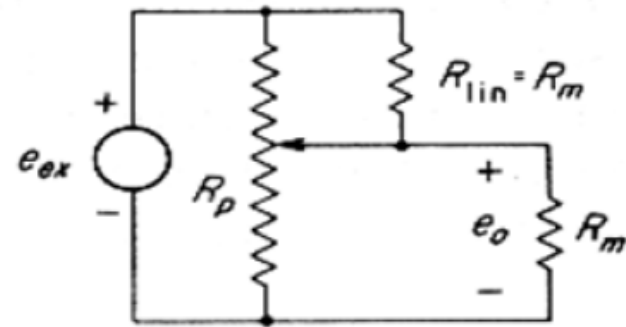
- Non-linearities as a result of **loading effects** 负载效应
- Resolution due to limited number of turns per unit distance

Loading Effect in Resistive Displacement Sensors

Loading effect



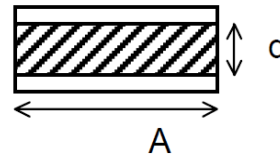
Compensation of loading effect



Capacitive Displacement Sensors

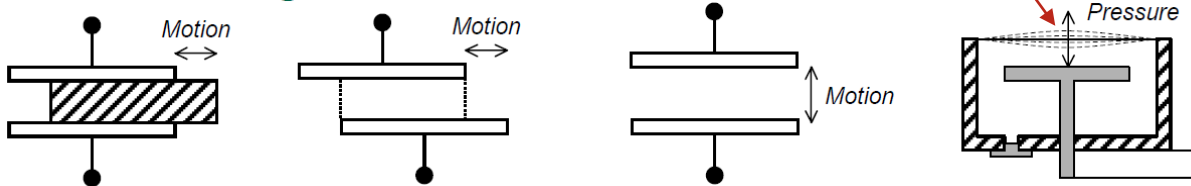
- **Principle:** The capacitance of a **parallel plate** capacitor is

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$



where d is the separation between the plates, A is the area of the plates, ϵ_0 is the permittivity of air and ϵ_r is the relative permittivity of the dielectric.

- **Sensor design:** A moving object is attached to the dielectric or the plates to generate capacitance changes

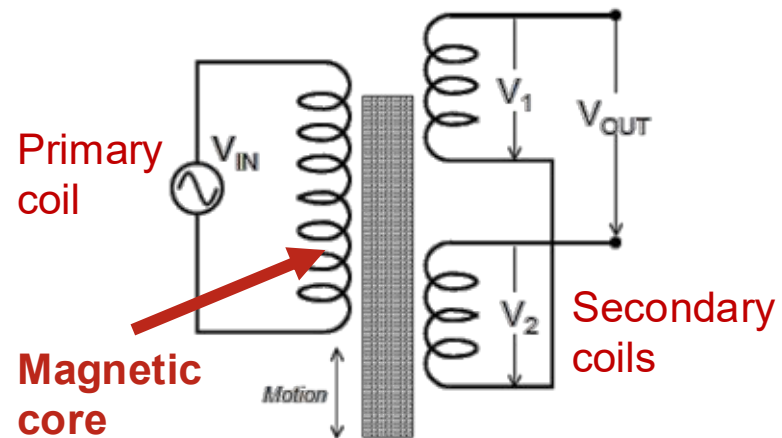


- High sensitivity: up to a few millimeters
- Can also be used to measure temperature and humidity (specially if the dielectric is air)
- Also commonly used to measure pressure: **microphones**

Inductive Displacement Sensors

- Linear Variable Differential Transformer (LVDT) 线性可变差动变压器
 - **Principle:** Motion of a magnetic core changes the **mutual inductance** of two **secondary coils** relative to a **primary coil**

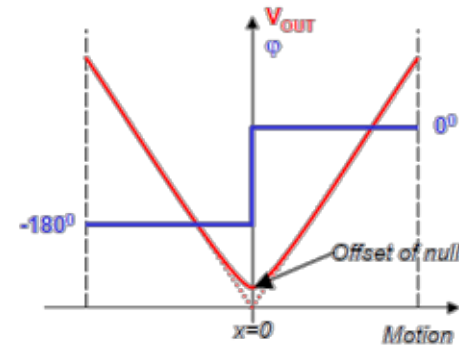
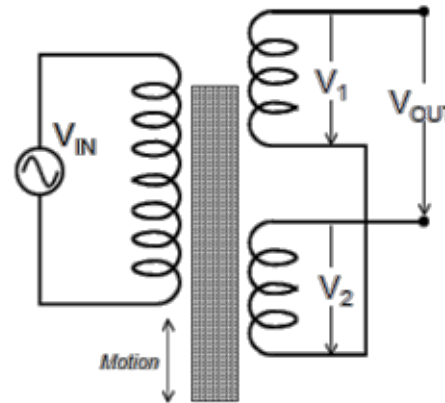
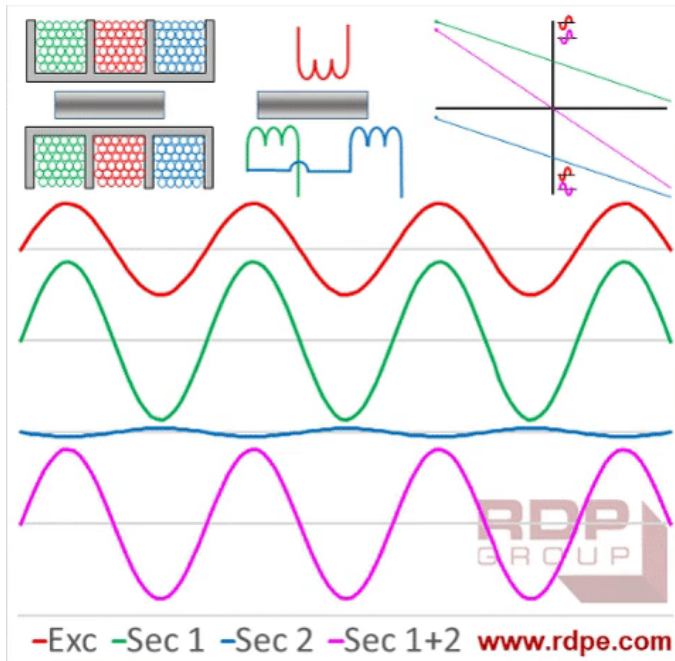
Sensor design:



- Primary coil voltage: $V_s \sin(\omega t)$
- Secondary coil induced voltage: $V_1 = k_1 \sin(\omega t + \phi)$ and $V_2 = k_2 \sin(\omega t + \phi)$ where k_1 and k_2 depend on the amount of coupling between the primary and the secondary coils, which is proportional to the position of the coil.

Inductive Displacement Sensors

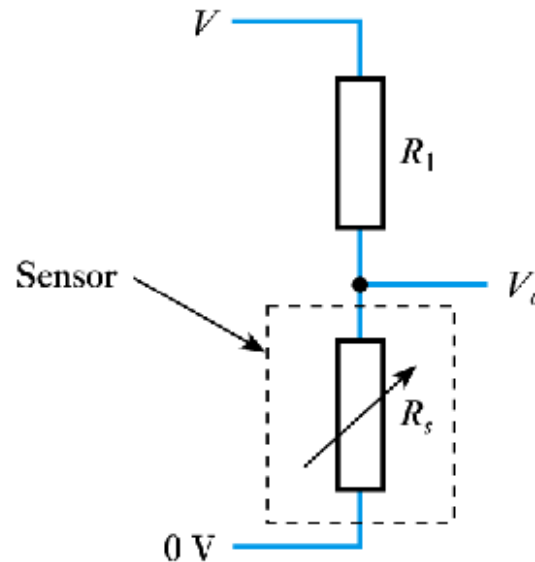
- Linear Variable Differential Transformer (LVDT)
 - Central position: $k_1 = k_2 \Rightarrow V_{OUT} = V_1 - V_2 = 0$
 - Displaced x units: $k_1 \neq k_2 \Rightarrow V_{OUT} = (k_1 - k_2)\sin(\omega t + \phi)$
 - Positive or negative displacements are determined from the phase of V_{OUT}



- LVDTs can measure from mm down to μm
- No mechanical wear ensures a long life
- Complete electrical isolation
- Absolute position output: always know its location

Exercise

- In the following circuit, if the resistance of the sensor varies linearly with the measured quantity, then the output voltage V_o is also linearly related to the measured quantity. True or False?



False

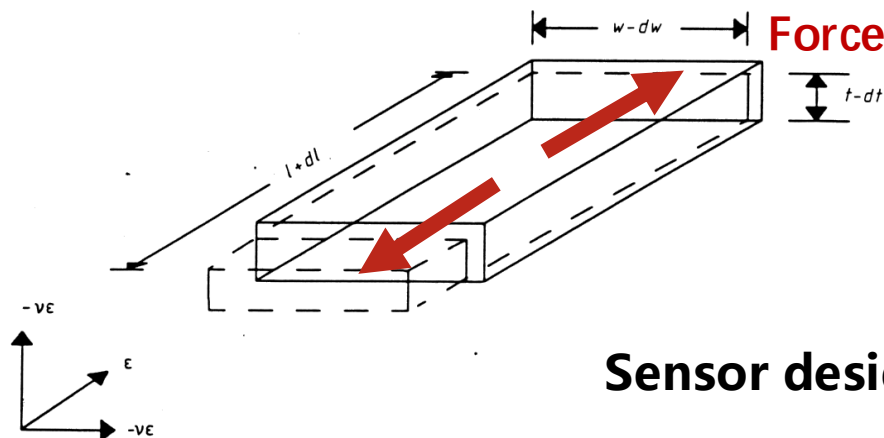
元件本身线性 不等于 系统线性

Strain Gauges

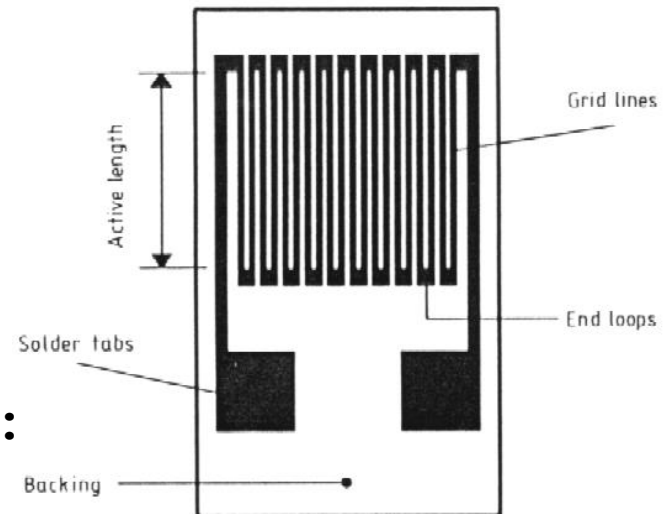
- **Principle: Piezo-resistive effect** - Resistance changes with stress.
- The resistance R of a strip of material of length L , cross-sectional area A and resistivity ρ is

$$R = \rho L / A$$

- Metal foil gauges: R change mainly because of the **geometric shape**.
- Semiconductor gauges: R change mainly because of the **piezo-resistivity**.



Sensor design:



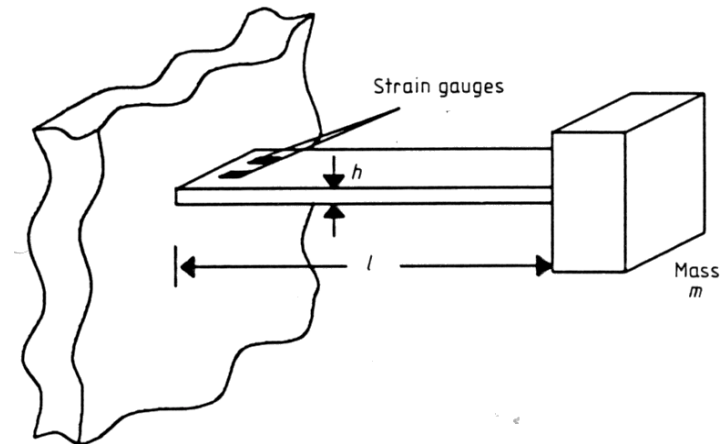
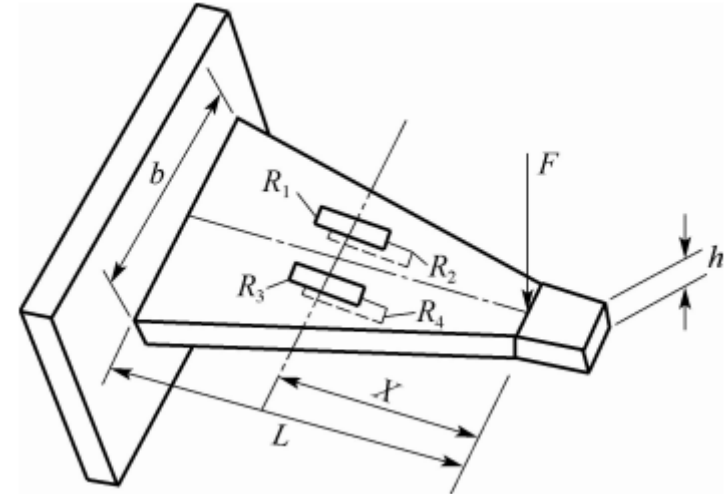
Force and Acceleration Sensors

- **Force sensors**

- Both compressive and tensile strains that can be easily measured in a **bridge arrangement**.

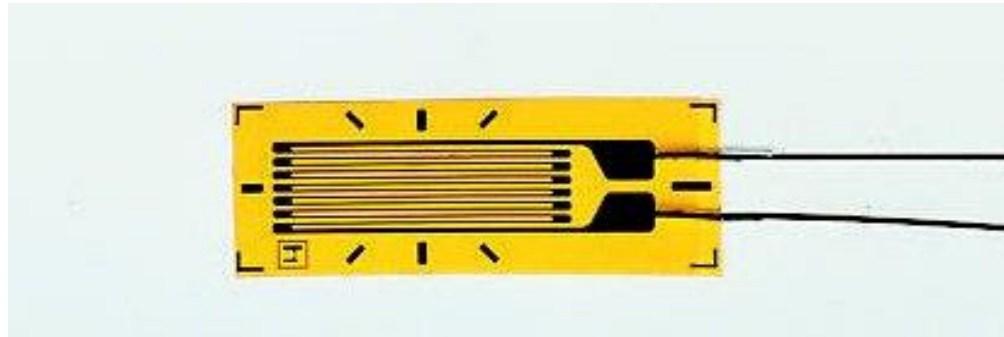
- **Acceleration sensors**

- Spring-mass-damper accelerometer: Covered previously
- Cantilever-beam with strain gauges
 - Mass is attached to the end of the cantilever
 - To reduce dampening: viscous fluids or permanent magnets around the sensor



Exercise

- The image below shows a strain gauge. With the device oriented as shown, what is the direction of sensitivity of the device?



- a) Perpendicular to the plane of the device.
- b) Vertical.
- c) Horizontal.

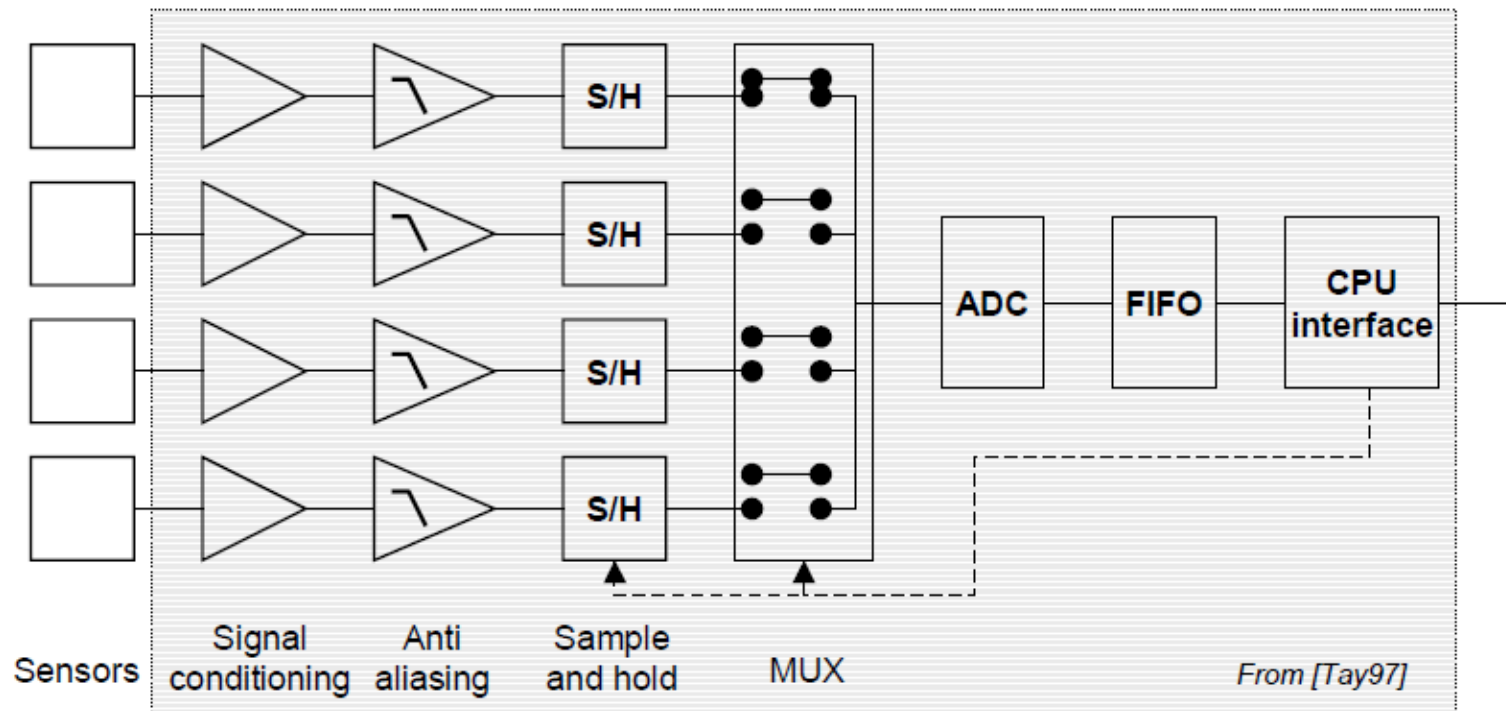
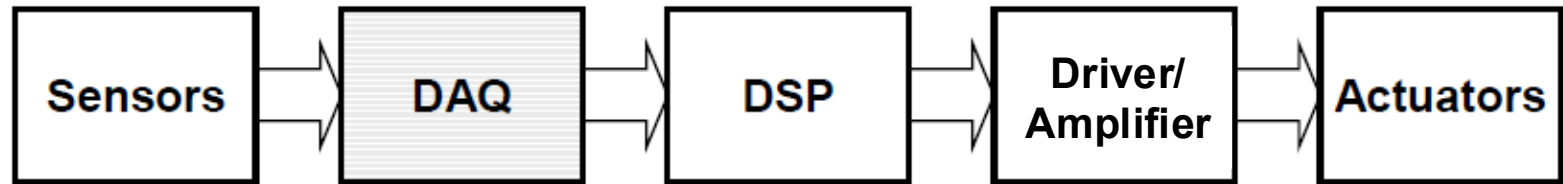
Summary – Sensing Principles: Mechanical Measurands

- Mechanical sensors to measure displacement, force, and acceleration
 - Resistive sensors
 - Capacitive sensors
 - Inductive sensors
 - Strain gauges and Cantilever beam-based sensors

Understanding the physics behind the sensor provides deeper insights than its readings alone.

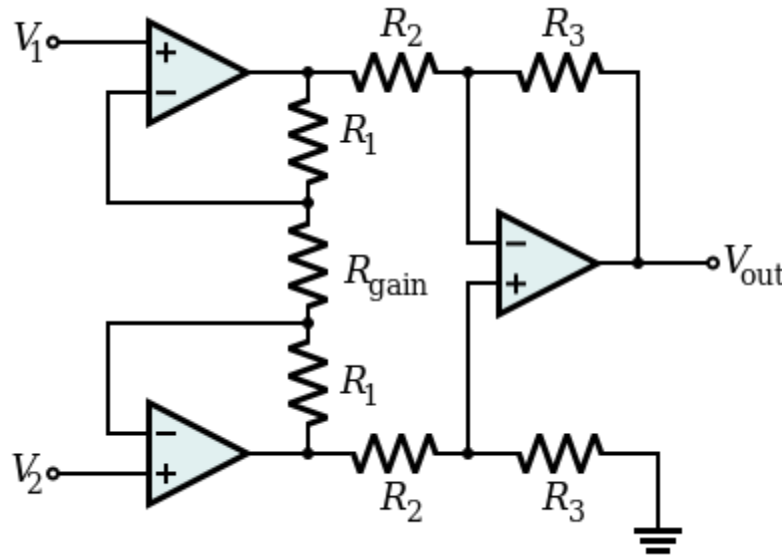
Data Acquisition

- Architecture of data acquisition systems



Signal Conditioning: Instrumentation Amplifiers

- Provide a **large amount of gain** for very low-level signals.

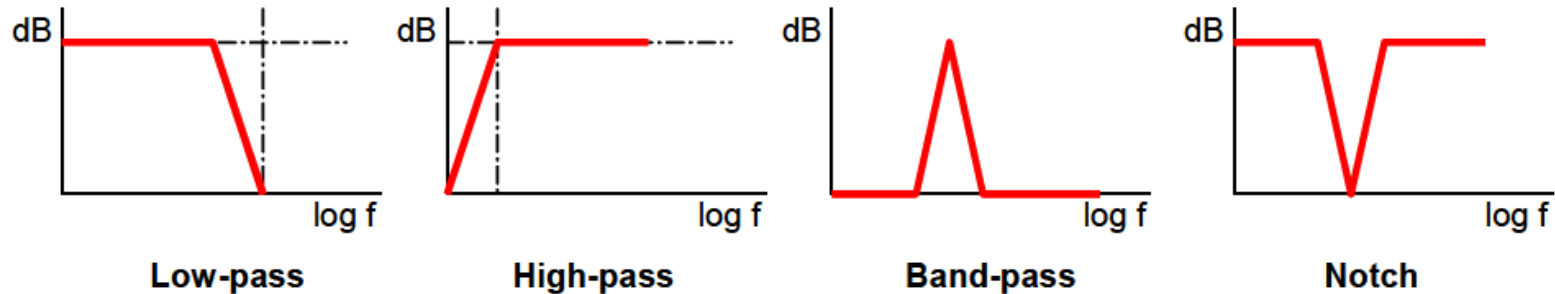


Typical instrumentation amplifier schematic

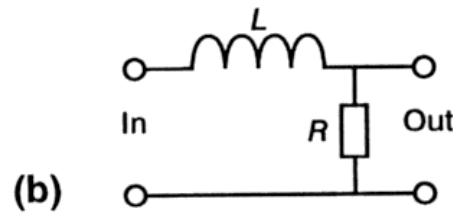
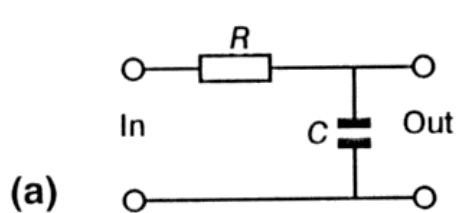
$$\text{Gain } A_v = \frac{V_{out}}{V_2 - V_1} = \left(1 + \frac{2R_1}{R_{gain}} \right) \frac{R_3}{R_2}$$

Signal Conditioning: Filters

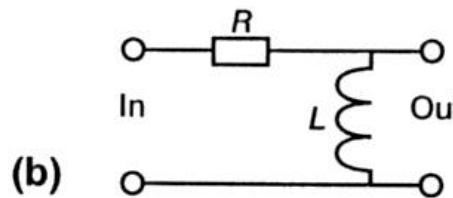
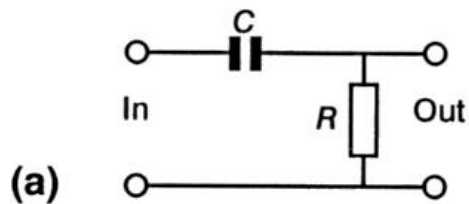
- Remove **unwanted bandwidths** from a signal.



- Low- and high-pass filters?



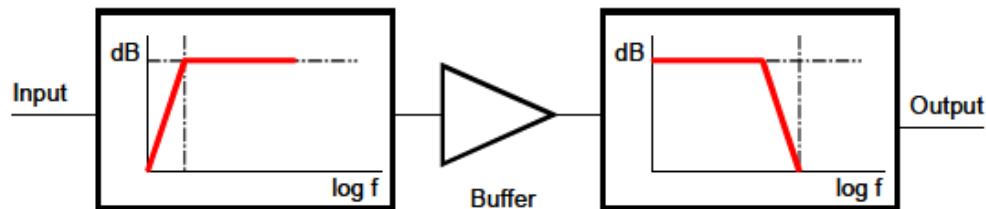
Low pass filters



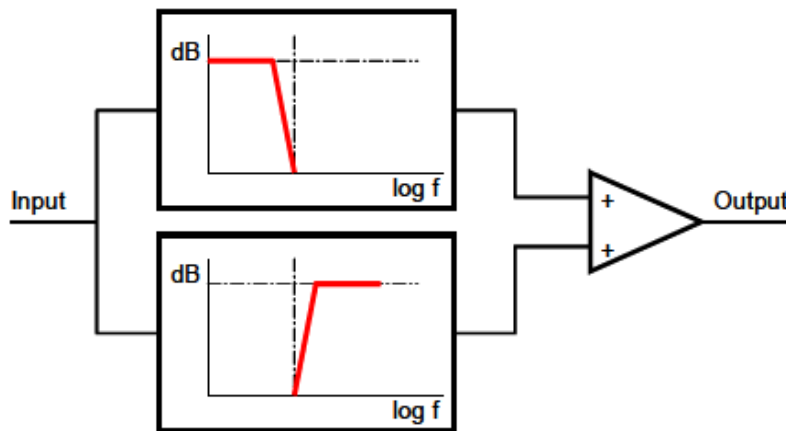
High pass filters

Signal Conditioning: Filters

- How to design a band-pass/band-stop filter?
 - High-pass and low pass **in series**: band-pass

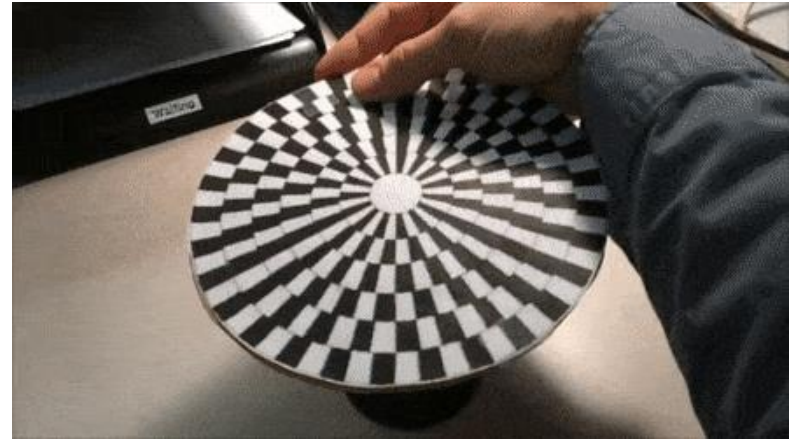
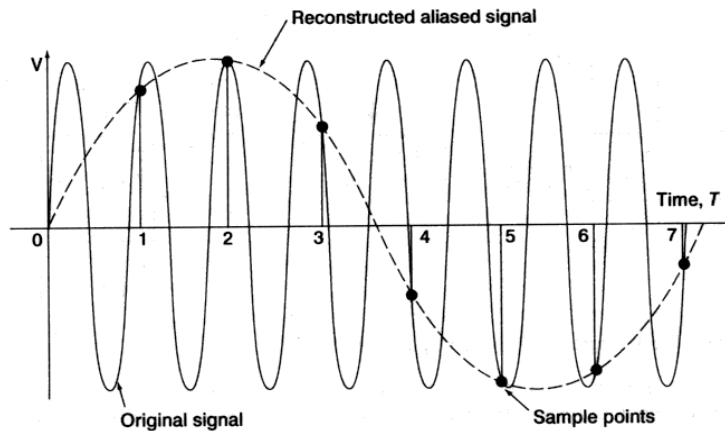


- High-pass and low-pass **in parallel** followed by a summer: band-stop



Anti-aliasing

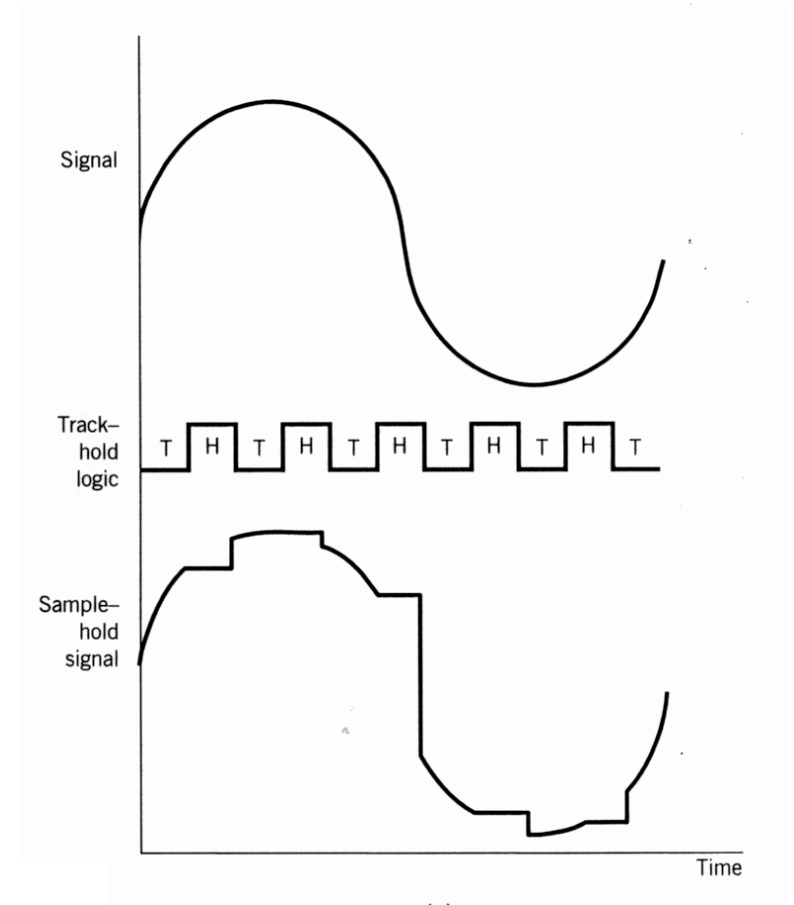
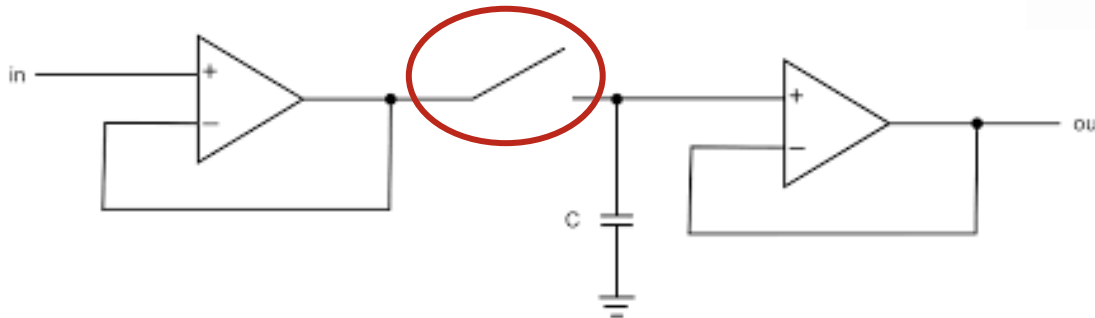
- Why do we need an anti-aliasing filter?
What would happen if the high frequency part is sampled directly?



- Once the signal has been sampled, the original high-frequency information is lost.
- Anti-aliasing filter 在采样前取出超过奈奎斯特频率的高频部分
 - A low-pass filter placed before the sample-and-hold circuit to prevent aliasing during sampling.

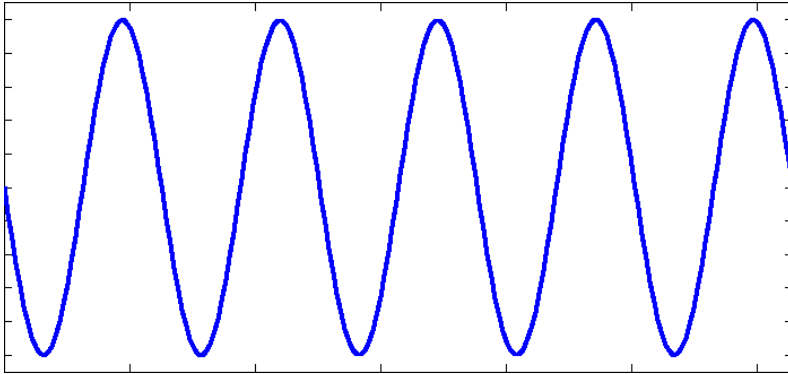
Sample and Hold

- A Sample and hold (S/H) circuit has two basic operating modes:
 - **Sample mode:** The output follows the input
 - **Hold mode:** The output is held constant until sample mode is resumed
- Why need S/H circuit?
 - To keep the signal constant during ADC conversion.



Sample and Hold

- Multimeter: hold for your eyes

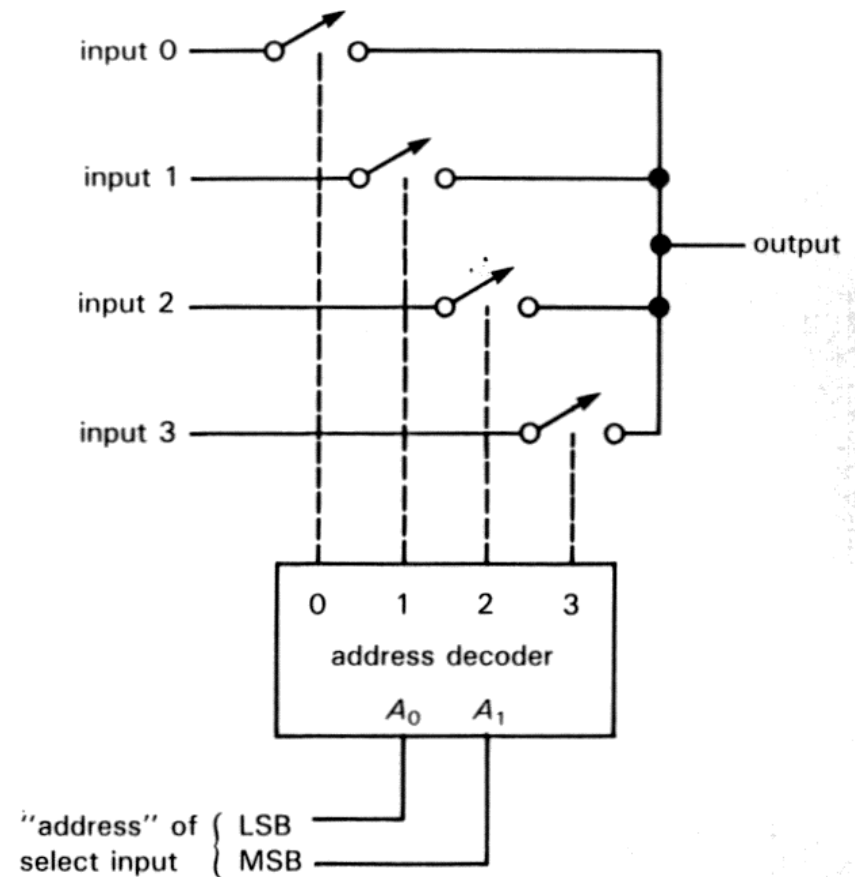


Multiplexers

- A multiplexer is a circuit that allows you to select any of several inputs, as specified by digital control signals.

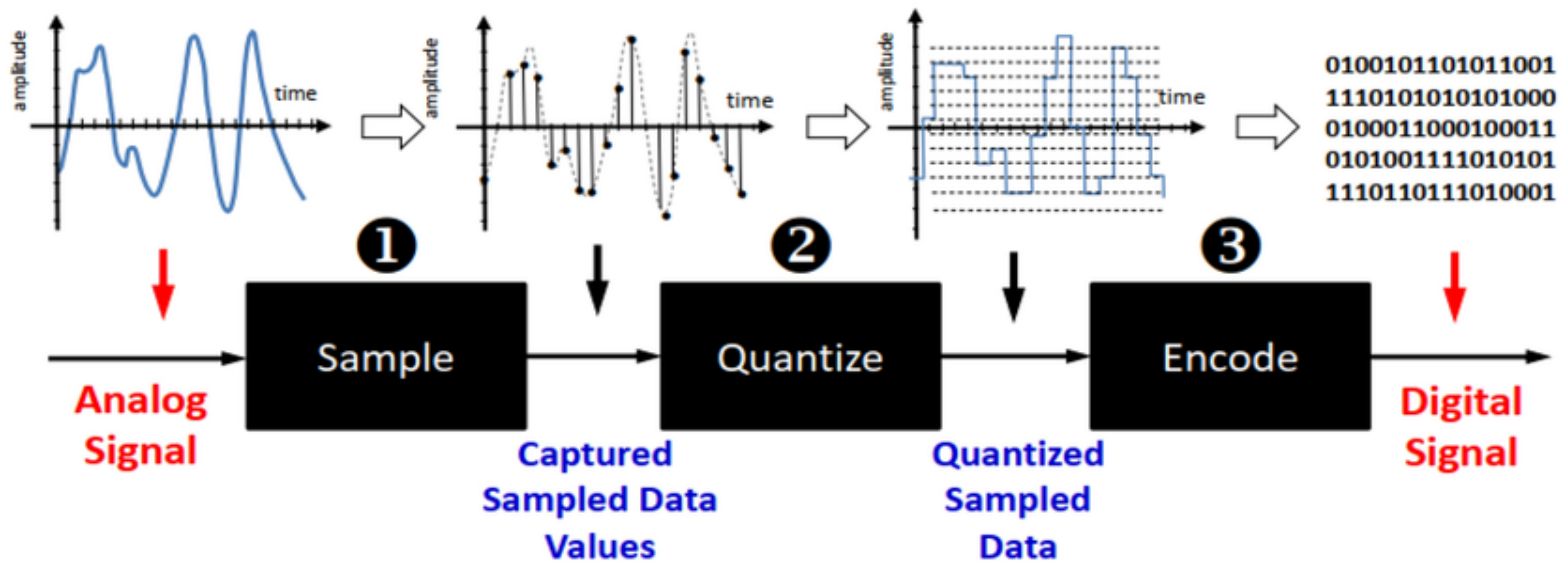
- How many selection bits are needed for a 4-channel multiplexer?

2



Analog to Digital Conversion (ADC)

Steps in ADC:



- **Features to take note:**

- Sampling rate
- Number of quantization levels
- Resolution
- Conversion time

Analog to Digital Conversion: Exercise

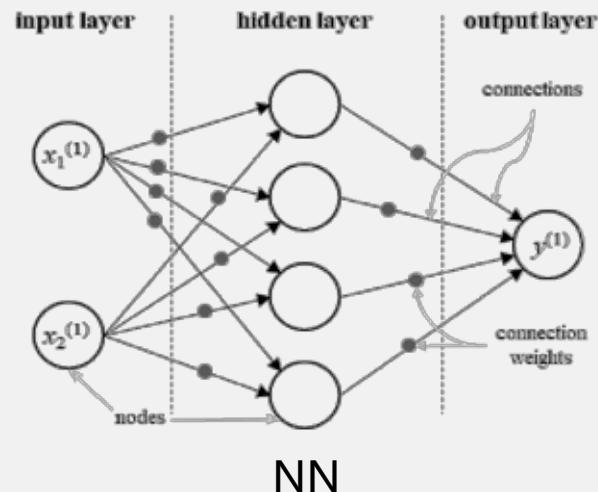
- The ADC has a 16-bit capacity, and full-scale range of 60 V.

Determine:

1. Number of quantization levels? $(L=2^{16}=65536)$
2. Resolution? $(\Delta V = \frac{60}{65536}) = (0.0009155V)$
3. Quantization error? Half of Resolution

Engineering Practice: NN-LUT Conversion

- Problem Statement: Your company developed a smart sensor which is equipped with a Neural Network (NN) model for data processing. The NN model has 2 input features and 1 output.
- Unfortunately, NN takes **too much space and computational effort** to be implemented on-chip.
- You are the engineer to optimize the on-chip NN model. One possible way is to **convert the NN to a Look-Up Table (LUT)**.



Output	Feature 1		
Feature 2			
		...	

LUT

Engineering Practice: NN-LUT Conversion

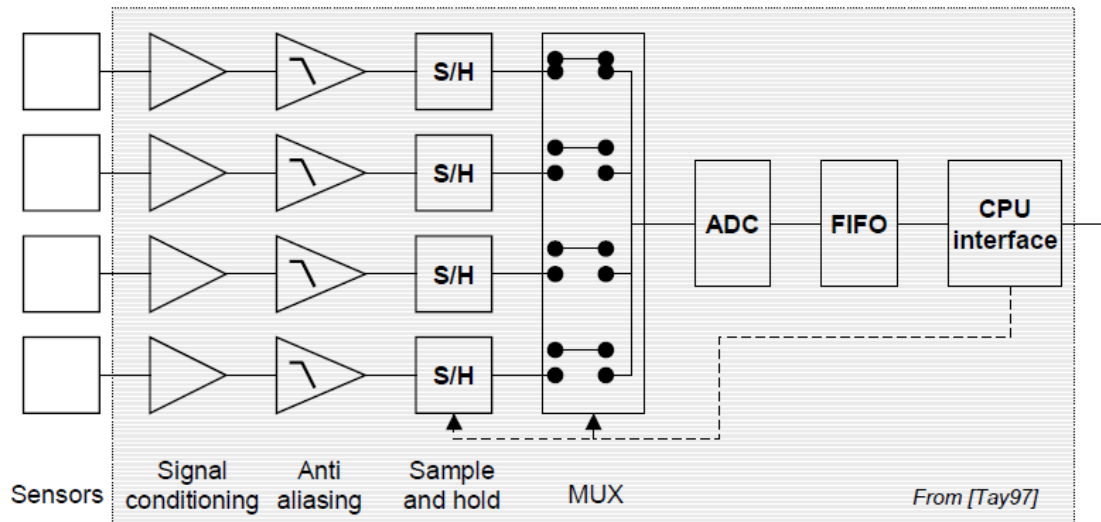
如果输入不在采样点上?
最近邻查找
插值

Engineering Practice: NN-LUT Conversion

Engineering Practice: NN-LUT Conversion

Summary – Data Acquisition

- Architecture of data acquisition systems
- Modules in data acquisition systems



References

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3. <https://www.decawave.com/technology1/>
4. [Chapter 3: Sensors \(pearsoned.co.uk\)](#)